

Project Status Report — Handover Document

Fractal Core Research · Elán Zainos Corona · Tlaxcala, México

Field	Value
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SSRN Preprint	https://ssrn.com/abstract=6418778
Zenodo DOI	https://doi.org/10.5281/zenodo.19446521
GitHub Repo	github.com/lnzainos/The-shadow-Node-Theory
Current Version	v29 — REAL CORPUS (721 verified cases)
Active Submission	MIT Golub Center for Finance and Policy — Oct 29-30, 2026
Report Date	June 2026

$\rho(t) = a \cdot t^b$

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Z-score: $b < 0$ = convergence · $b \approx 0$ = equilibrium · $b > 1$ = Roche Radius (absorption)

1. What Is Shadow Node Theory

Shadow Node Theory (SNT) is a formal, empirically-grounded model proposing that when two entities — a dominant hub and a peripheral node — interact over time, the ratio of their relative standing follows a power law $R(t) = a \cdot t^b$, where the exponent b (the satellization exponent) quantifies both the direction and velocity of the dominance dynamic. The model's central claim is that **institutional friction is the primary determinant of b** — and is now measurable across domains.

The theory is verified across 721 real empirical cases spanning 11 domains and 30 orders of magnitude of temporal scale — from digital platform churn (hours) to galactic merger timescales (billions of years). Every b value and R^2 is reproducible from public primary sources via published Python scripts.

Key Definitions

Term	Definition
Hub	Dominant node with greater accumulated advantage
Shadow Node	Peripheral entity whose productive output is systematically extracted
Satellization	The process by which a hub absorbs a shadow node over time
Exponent b	Satellization velocity. $b < 0$ =convergence, $b = 0$ =equilibrium, $b > 0$ =satellization, $b \geq 1$ =absorption
Roche Radius	$b = 1.0$ — critical threshold of accelerating absorption (astronomical analogy)
Leapfrog	Shadow node escapes satellization via orthogonal dimension (new competitive axis)
Trigger (abrupt)	Dateable event causing immediate satellization acceleration
Trigger (gradual)	Cumulative process with no single central event
ASI	Atomic Sovereignty Index: $ASI = \Delta H \times \alpha / F$ — measures cognitive/behavioral sovereignty
Institutional Friction	Aggregate structural resistance a system deploys against rapid absorption

2. Corpus Status — v28 vs v29

In June 2026, a full audit of the corpus revealed that v28 (502 cases) contained ~188 synthetic b values generated via `np.random.normal()` and R^2 values as low as -7.332 (statistically impossible). These were never published as final — all SSRN and Zenodo submissions marked the corpus as 'in progress'. The v29 reconstruction replaces all values with real, verifiable primary-source data.

Metric	v28 (OBSOLETE)	v29 (CURRENT — USE THIS)
Total cases	502	721
Data source	Synthetic (<code>np.random</code>)	100% real, verifiable
Statistically sig.	31%	89%
Corrupt R^2 values	~99	0
Spearman ρ (friction vs b)	-0.74 (inflated)	-0.68 (real, $n=714$)
Central Mann-Whitney	Not verifiable	$p = 2.4 \times 10^{-10}$ ■■■■

Traceable to source	NO	YES (public GitHub)
H-phi hypothesis	Untested	REFUTED (3 validations)
Status	HISTORICAL RECORD ONLY	■ ACTIVE — CITE THIS

3. v29 Corpus — Primary Sources & Key Results

Domain	Cases	Primary Source	Key Finding (b)
Historical city-pairs (A)	25	Historical population data	Abrupt triggers: $b \approx +0.72$ (Córdoba→Madrid $b = +1.79^{***}$)
Subnational — Mexico (C) 32		INEGI 2022 GDP per capita	Tlaxcala→CDMX $b = +0.17^*$, Querétaro $b = -0.14^*$ (leapfrog)
Subnational — USA (C)	10	US Census Bureau	Appalachia→NE $b = -0.08^{**}$ (convergence)
Sovereign countries (D)	446	Maddison Project Database 2020	Global mean $b \approx +0.09$, Europe $b \approx -0.007$, Sub-Saharan $b \approx +0.258$
Epidemic growth (E)	234	Johns Hopkins / Our World in Data	$b \approx 0.91$ — friction-free baseline
Predator-prey (B)	4	MacLulich 1937 / Elton & Nicholson	$b \approx 0.10$ — interdependence as brake
Planetary/stellar (P)	10	Open Exoplanet Catalogue	$b \approx +1.03$ — gravity as pure satellization
Digital platform (G)	4774 users	HackerEarth 2026	ASI: ROC-AUC=0.715, 5-Event Wall threshold

Three Core Statistical Findings (v29)

Finding	Result	Test
Abrupt triggers produce faster satellization than gradual ones	Ratio 5:1 actual vs. historical corpus	Mann-Whitney U=24,802, $p = 1.91 \times 10^{-11}$, n=486
Institutional friction is dominant predictor of friction-free baseline	Fictional feed demand High friction: $b \approx +0.89$	Spearman $\rho = -0.68$, $p = 2.5 \times 10^{-11}$, n=714
Political sovereignty = ecological interdependence	20% courts satellization predator-prey system	Mann-Whitney $p = 2.4 \times 10^{-11}$

N-Body Matrix Correction — Mexican National System

Standard binary models (Tlaxcala vs Puebla) underestimate Tlaxcala’s satellization gradient by **9.3×** because 89.2% of extraction flows toward CDMX, not Puebla. The N-body correction (32 federal entities, INEGI 2022) reveals a power-law distribution of satellization weights ($b = -0.473$, $R^2 = 0.838$, $p < 0.001$) consistent with preferential attachment predictions. Querétaro ($b = -0.155$) and Nuevo León ($b = -0.058$) document the first confirmed leapfrog cases within the national system — both via orthogonal dimension (aerospace and export manufacturing respectively).

Atomic Sovereignty Index (ASI)

$ASI = \delta H \times \alpha / F$ where δH = Shannon entropy of behavioral sequence, α = autonomy ratio (self-directed vs prompted actions), F = friction index. Applied to 4,774 HackerEarth users (409,287 events), ASI achieves held-out ROC-AUC = 0.715 using exclusively first-session features. The 5-Event Wall is the activation threshold. The dominant retention predictor is AI agent adoption — interpreted as cognitive leapfrog.

4. Publication Track & Submission History

Version / Target	Date	Status	Notes
SSRN Preprint v1.0 (Zenodo DOI: 10.5281/zenodo.6913132)	Early 2026	■ LIVE	v2.0, Spanish+English abstract, 5 domains
Oxford / Communications in Nonlinear Science (COMNET)	June 2026	■ REJECTED	502-case version (v28). Rejection reason not disclosed. Do NOT CITE
PLOS ONE — SNT_paper_PLOS_721cases_v5_DRAFT	Jan 2026	■ DRAFT	721 real cases, v29 corpus. Ready for submission review.
MIT Golub Center for Finance and Policy (GCFP)	June 2026 submitted	■ UNDER REVIEW	13th Annual Conference, Oct 29-30, 2026, Cambridge MA. Rejection reason not disclosed.
GitHub repo: github.com/lnzainos/The-shadow-node-theory	Ongoing	■ PUBLIC	README v29, replication scripts, full corpus CSV

MIT GCFP Paper — Key Reframe

The MIT submission reframes SNT specifically for financial regulation audiences. The satellization exponent b becomes a **regulatory diagnostic metric**: does a regulatory environment hold peripheral economies near $b \approx 0$ (stable orbit), push them below $b < 0$ (convergence), or allow drift toward $b > 1$ (absorption)? European country pairs under common market institutions show $b \approx -0.007$. Sub-Saharan African pairs show $b \approx +0.258$. The paper provides a four-step diagnostic framework for applying this to regulatory assessment, trade agreement evaluation, and financial integration policy.

5. Drive File Inventory (Folder: /Snt)

File	Type	Status	Purpose
SNT_MIT_GCFP_2026 — Institutional For Google Doc	Google Doc	■ ACTIVE	MIT submission — current canonical paper
SNT_paper_PLOS_721cases_v5_DRAFT	Google Doc	■ DRAFT	PLOS ONE version with 721 cases
ARCHIVO_snt_oxford_submission_COMNET-RECH-ZARCH	Google Doc	■ ARCHIVE	Oxford rejected version — DO NOT CITE
snt_oxford_submission_final	Google Doc	■ ARCHIVE	Duplicate of Oxford version
SNT_theoretical_framework_v29_EN	Google Doc	■ ACTIVE	Full theoretical framework v29 in English
SNT_marco_teorico_v29_ES	Google Doc	■ ACTIVE	Full theoretical framework v29 in Spanish
README_v29_para_GitHub	Google Doc	■ ACTIVE	GitHub README — copy to repo
snt_paper_investigacion_economica.docx	DOCX	■ ACTIVE	Spanish econ paper — SNT v2.2, 502-case (v28)
shadow_node_theory_SSRN_v10.docx	DOCX	■ ACTIVE	SSRN bilingual version
marco_teorico_v03.docx	DOCX	■ OLDER	Earlier theoretical framework
snt_corpus_final.csv	CSV	■ ACTIVE	Main corpus (check if v28 or v29 values)
shadow_node_resultados_v2.csv	CSV	■ ACTIVE	4 original historical cases with full stats
snt_v2_vectores.csv	CSV	■ ACTIVE	8 Mexican states — b vectors 1940-2022
shadow_node_maddison_resumen.csv	CSV	■ ACTIVE	Maddison 4-case summary
snt_corpus_50_analisis.py	Python	■ ACTIVE	Main analysis script
snt_corpus_50_analisis_v2.py	Python	■ ACTIVE	Updated analysis script

snt_asl_operacionalizacion.py	Python	■ ACTIVE	ASl operationalization script
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6. Falsifiability Criteria (RC1–RC8)

The model includes explicit refutability criteria — a key strength for peer review. All 8 criteria have been tested against the v29 corpus.

RC	Refutation Condition	v29 Status
RC1	Power law fits no better than linear/exponential across all domains	■ NOT REFUTED — power law wins in low-friction domains
RC2	b is not reproducible from primary series	■ NOT REFUTED — full replication package published
RC3	Abrupt triggers produce same b as gradual (no velocity difference)	■ NOT REFUTED — 5.9× ratio, $p=1.91\times 10^{-5}$ ■■■
RC4	Friction index is not correlated with b	■ NOT REFUTED — $\rho=-0.68$, $p=2.5\times 10^{-9}$ ■■■
RC5	N-body matrix does not change satellization estimates	■ NOT REFUTED — 9.3× underestimate confirmed
RC6	Shadow node reverses satellization from inside without exogenous triggers	■ NOT REFUTED — all leapfrog cases required external triggers
RC7	ASI does not predict behavioral outcomes better than chance	■ NOT REFUTED — ROC-AUC=0.715 (vs 0.5 chance)
RC8	Mutual interdependence does not brake satellization	■ NOT REFUTED — predator-prey = sovereign country equivalence

7. Next Steps

Priority	Action	Detail
P0 — Immediate	MIT response monitoring	Conference Oct 29-30 2026. Prepare presentation slides if accepted. Check SSRN for revision
P0 — Immediate	GitHub README update	Push README_v29_para_GitHub to repo. Update corpus CSV to v29 values. Tag release v29
P1 — Pre-submission	PLOS ONE final review	SNT_paper_PLOS_721cases_v5_DRAFT needs one pass: verify all 721 b values cite primary
P1 — Pre-submission	Corpus CSV final audit	Verify snt_corpus_final.csv has zero synthetic values. Check R² column: must be in [0,1]. Run
P2 — Research	Genomic SNT cross-domain	The hub-satellite topology of the economic model is now being tested in genomics (SNT Gen
P2 — Research	Sentinel Omega SNT integration	Sentinel Omega uses SNT topology for geophysical prediction. A methodological paper linking
P3 — Publication	Unified theory paper	Marco Teórico Unificado — combines SNT economics + genomics + geophysics + Fractal C

8. Quick Reference for New Instance

When continuing this project in a new conversation, paste this context block:

PROJECT: Shadow Node Theory (SNT) — Fractal Core Research
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CURRENT VERSION: v29 — 721 REAL cases (v28/502 cases = OBSOLETE, synthetic data)
ACTIVE SUBMISSION: MIT GCFP Oct 2026 (institutional friction as regulatory variable)
DRAFT READY: PLOS ONE (SNT_paper_PLOS_721cases_v5_DRAFT)

CORE FORMULA: $R(t) = a \cdot t^b$ | b=satellization exponent
FINDING 1: Abrupt triggers 5.9x faster than gradual ($p=1.91e-5$)
FINDING 2: Institutional friction predicts b ($\rho=-0.68$, $p=2.5e-97$, $n=714$)

FINDING 3: b varies 0-1 (friction) to +0.91 (epidemic, no friction)
ROCHE RADIUS: $b=1.0$ = point of accelerating absorption
ASI: $\Delta H \times \alpha / F$ | ROC-AUC=0.715 (HackerEarth 4774 users)
DO NOT CITE: v28 figures, snt_oxford_submission, 502-case corpus